NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL

Department of Mathematics

CPDS

Assignment - II

Course: CPDS

Instructor: Dr. Sivalingam S M

1. Given a list of integers and a target S, find two distinct positions $i < j$ such that $a_i + a_j = S$. Return the two indices, or -1 if no such pair exists. If more than one pair exists, any one is acceptable.

   (a) Write a solution using two nested loops.

   (b) Write another solution using a dictionary to record previously visited elements.

   (c) Test both solutions when the list contains repeated elements and when no pair exists.

   Example. For `[2, 8, 4, 11, 5]` and $S = 9$, a valid answer is `(2, 4)` (indices begin at 0).

2. A string contains only the symbols `( ) [ ]` { }. It is balanced if every opening bracket has a matching closing bracket of the same type, in the correct order.

   (a) Use a Python list as a stack to check whether a given string is balanced.

   (b) Show the contents of the stack after reading each character of the first example.

   (c) Test the function for an empty string, an unmatched opening bracket and an unmatched closing bracket.

   Example. `([{}])` is balanced, whereas `([)]` is not.

3. A contiguous sublist consists of consecutive elements of a list. For a nonempty list of integers, find the largest possible sum of a nonempty contiguous sublist.

   (a) Solve the problem by considering all possible starting and ending positions.

   (b) Find a method that scans the list once, updating the best sum ending at the current position.

   (c) Modify the second method to also return the starting and ending indices of a sublist attaining the maximum sum. Test a list containing only negative numbers.

   Example. For `[-2, 1, -3, 4, -1, 2, 1, -5, 4]`, the maximum sum is 6, attained by `[4, -1, 2, 1]`.

4. Two words are anagrams if they contain the same letters with the same frequencies, possibly in a different order. Assume that all words contain only lowercase English letters.

   (a) Write a function to check whether two given words are anagrams.

   (b) Given a list of words, group together those that are anagrams of one another. Use a dictionary to store the groups.

   (c) Test your function with repeated words and with words that have no anagrams in the list.

   Example. For `['eat','tea','tan','ate','nat','bat']`, the groups are `['eat','tea','ate']`, `['tan','nat']` and `['bat']`, in any order.

5. Given a list a of integers and an integer k with $1 \le k \le \text{len}(a)$, find the maximum sum of exactly k consecutive elements.

(a) Write a direct solution that calculates the sum of every block of k elements.

(b) Write another solution that obtains the next block sum by subtracting the element leaving the block and adding the element entering it.

(c) Return the maximum sum and its starting index. Test $k = 1$ and $k = \text{len}(a)$.

Example. For `[2, 1, 5, 1, 3, 2]` and $k = 3$, the maximum sum is 9, from `[5, 1, 3]`.

6. A matrix is represented by a rectangular nested Python list. Read its elements clockwise in a spiral, starting at the top-left corner and moving along the top row first. Do not use NumPy.

(a) Write a function that returns the elements in spiral order.

(b) Explain how the top, bottom, left and right boundaries change after each pass.

(c) Test your function on a single-row matrix and a single-column matrix.

Example. For $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}$, the output is `[1,2,3,4,8,12,11,10,9,5,6,7]`.

7. In postfix notation, an operator follows its two operands. For example, `3 4 +` represents $3 + 4$. An expression is supplied as a list of string tokens containing integers and the operators `+`, `-`, `*`. Assume the expression is valid.

(a) Use a Python list as a stack to evaluate a postfix expression.

(b) Trace the stack after processing each token of the example. Pay attention to the order of operands in subtraction.

(c) Test the function with a single integer and with an expression containing negative integers.

Example. `['5','1','2','+','4','*','+','3','-']` evaluates to 14.

8. There are n disks of different sizes on peg A, arranged with the largest at the bottom. Move all disks to peg C, using peg B as an auxiliary peg. Only one disk may be moved at a time, and a larger disk may never be placed on a smaller disk.

(a) Write a recursive function that prints the moves needed to transfer n disks from one peg to another.

(b) List the moves for $n = 3$ and check that the rules are followed after every move.

(c) Count the moves made by the program for $n = 1, 2, 3, 4$. Find a formula for the minimum number of moves.

Example. For $n = 3$, the minimum number of moves is 7.